

SYDNEY TECHNICAL HIGH SCHOOL

Celebrating 100 years of public education



Mathematics Extension 2

HSC ASSESSMENT TASK

JUNE 2011

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions

NAME : _____

TEACHER : _____

QUESTION 1	QUESTION 2	QUESTION 3	TOTAL

Question 1 (17 marks)

a) Find $\int x^3 \ln x \, dx$ 3

b) Show that $\int_5^6 \frac{x+2}{(x-4)(x-1)} \, dx = \ln \frac{16}{5}$ 3

c) $1 + i$ and $3 - i$ are zeroes of a monic polynomial, $P(x)$, of degree 4 with real coefficients.
Express $P(x)$ as a product of two real quadratic factors. 2

d) i) Solve the equation $16x^4 - 16x^2 + 1 = 0$ 2
using the identity $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.

ii) Solve the equation $16x^4 - 16x^2 + 1 = 0$ as a quadratic in x^2 , 3
and hence find the exact value of $\cos \frac{\pi}{12}$. Justify your answer.

e) The equation $x^3 + 3px + q = 0$ has a double root at $x = k$.

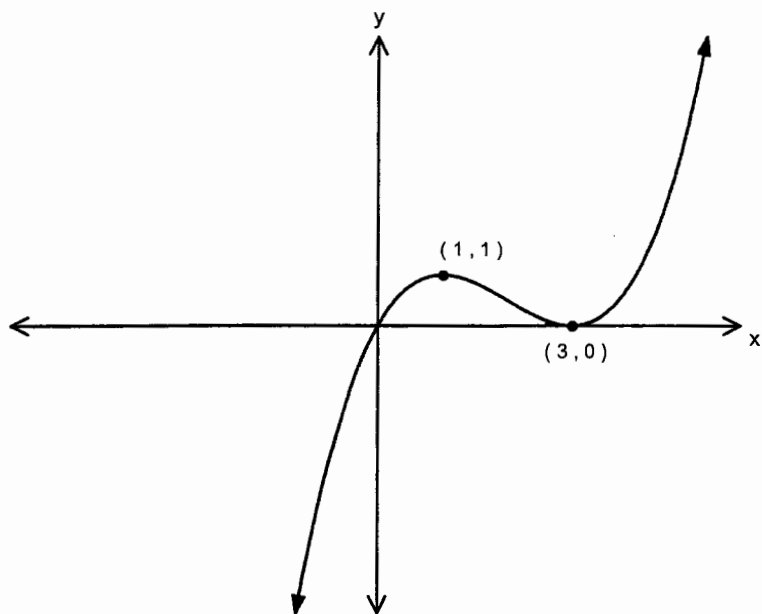
i) Show that $p = -k^2$ 1

ii) Hence solve $x^3 - 6ix + 4 - 4i = 0$ given that it has a double root. 3

Question 2 (17 marks) - Start a new page

a) Below is a sketch of the function $y = g(x)$.

There are stationary points at $(1, 1)$ and $(3, 0)$.



On separate diagrams, draw neat sketches of the following, clearly showing any important features or points.

i) $y = \int g(x) dx$ 2

ii) $y = \frac{1}{g(x+1)}$ 2

iii) $y^2 = g(x)$ 2

iv) $y = \cos^{-1} g(x)$ 2

Question 2 (continued)

b) The points $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ lie on the hyperbola $xy = 4$.

i) What is the eccentricity of the hyperbola $xy = 4$. 1

ii) Find the equation of the chord PQ . 2

iii) Given that the chord PQ always passes through the point $(4, 2)$ 1
show that $pq = p + q - 2$.

iv) Find the locus of M , the midpoint of PQ . 2

c) Evaluate $\int_0^{\frac{\pi}{6}} \sin 2x \sin 4x \, dx$ 3

Question 3 (17 marks) - Start a new page

a) Find $\int \frac{2x}{x^2+10x+26} dx$ 3

b) Find $\int \frac{1+\sin x}{\cos^2 x} dx$ 2

c) The equation $2x^4 - 3x^2 - 2x + k = 0$ has a triple root. 3
Find the value of k .

d) If α, β and γ are the roots of the equation $x^3 - 6x^2 - 4x + 2 = 0$.

i) Find the monic polynomial equation with roots $2\alpha, 2\beta, 2\gamma$ 2

ii) Find the constant term in the monic polynomial with roots

$\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$ 3

e) Use integration by parts to evaluate $\int_0^{\infty} 3y e^{-y} (1 - e^{-y})^2 dy$ 4

End of paper. ☺

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

EXTENSION 2 SOLUTIONS JUNE 2011

QUESTION 1

a) $\int x^3 \ln x \, dx$ $u = \ln x$ $u' = \frac{1}{x}$
 $v = \frac{1}{4} x^4$ $v' = x^3$

$$= \frac{1}{4} x^4 \ln x - \int \frac{1}{4} x^3 \, dx$$

$$= \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + C$$

b) $\int_5^6 \frac{x+2}{(x-4)(x-1)} \, dx$ $\frac{x+2}{(x-4)(x-1)} = \frac{A}{x-4} + \frac{B}{x-1}$

$$= \int_5^6 \frac{2}{x-4} - \frac{1}{x-1} \, dx$$

$\therefore x+2 = A(x-1) + B(x-4)$
 $x=1 \Rightarrow 3 = -3B \quad \therefore B = -1$

$A+B=1 \quad \therefore A=2$

$$= 2 \ln(x-4) - \ln(x-1) \Big|_5^6$$

$$= (2 \ln 2 - \ln 5) - (2 \ln 1 - \ln 4)$$

$$= 2 \ln 4 - \ln 5$$

$$= \ln 16 - \ln 5$$

$$= \ln \frac{16}{5}$$

c) roots are $1+i, 1-i, 3-i, 3+i$

Sum 2 6

product 2 10

$$\therefore P(x) = (x^2 - 2x + 2)(x^2 - 6x + 10)$$

$$d) \quad i) \quad 16x^4 - 16x^2 + 1 = 0$$

$$2(8x^4 - 8x^2 + 1) - 1 = 0$$

$$\text{let } x = \cos \theta$$

$$2(8\cos^4\theta - 8\cos^2\theta + 1) - 1 = 0$$

$$2\cos 4\theta = 1$$

$$\cos 4\theta = \frac{1}{2}$$

$$4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

$$\therefore x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}$$

$$ii) \quad 16x^4 - 16x^2 + 1 = 0$$

$$\text{let } u = x^2$$

$$16u^2 - 16u + 1 = 0$$

$$u = \frac{16 \pm \sqrt{16^2 - 4 \cdot 16 \cdot 1}}{32}$$

$$= \frac{16 \pm \sqrt{192}}{32}$$

$$= \frac{16 \pm 8\sqrt{3}}{32}$$

$$= \frac{2 \pm \sqrt{3}}{4}$$

$$\therefore x^2 = \frac{2 \pm \sqrt{3}}{4}$$

$$x = \pm \frac{\sqrt{2 \pm \sqrt{3}}}{2}$$

$$\therefore \cos \frac{\pi}{12} = \frac{\sqrt{2 + \sqrt{3}}}{2} \quad (\text{largest positive})$$

$$e) \quad 1) \quad x^3 + 3px + q = 0$$

$$3x^2 + 3p = 0$$

$$x^2 = -p$$

double root of $x = k$

$$\Rightarrow k^2 = -p$$

$$p = -k^2$$

$$ii) \quad \text{roots } k, k, \alpha$$

$$2k + \alpha = 0$$

$$k^2 + 2k\alpha = -6i$$

$$k^2\alpha = -4 + 4i$$

$$\text{but } k^2 = 2i$$

$$\alpha(2i) = -4 + 4i$$

$$\alpha = \frac{-4 + 4i}{2i} \times \frac{i}{i}$$

$$= \frac{-4i - 4}{-2}$$

$$= 2i + 2$$

$$\therefore 2k + 2i + 2 = 0$$

$$2k = -2i - 2$$

$$k = -i - 1$$

$\therefore -i-1, -i-1, 2i+2$ are solutions

$$x^3 + 3px + q = x^3 - 6ix + 4 - 4i$$

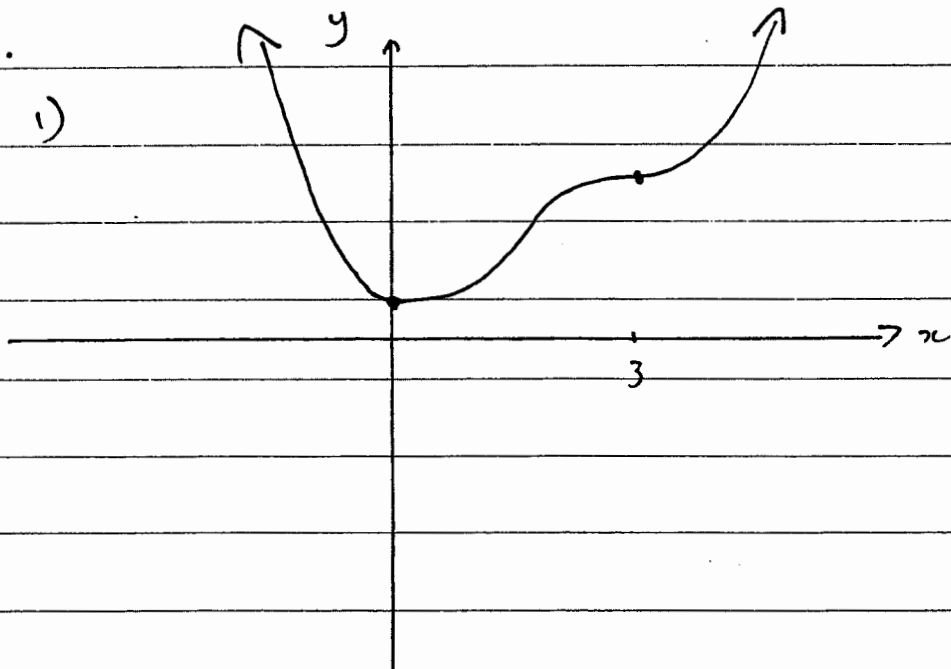
$$3p = -6i$$

$$q = 4 - 4i$$

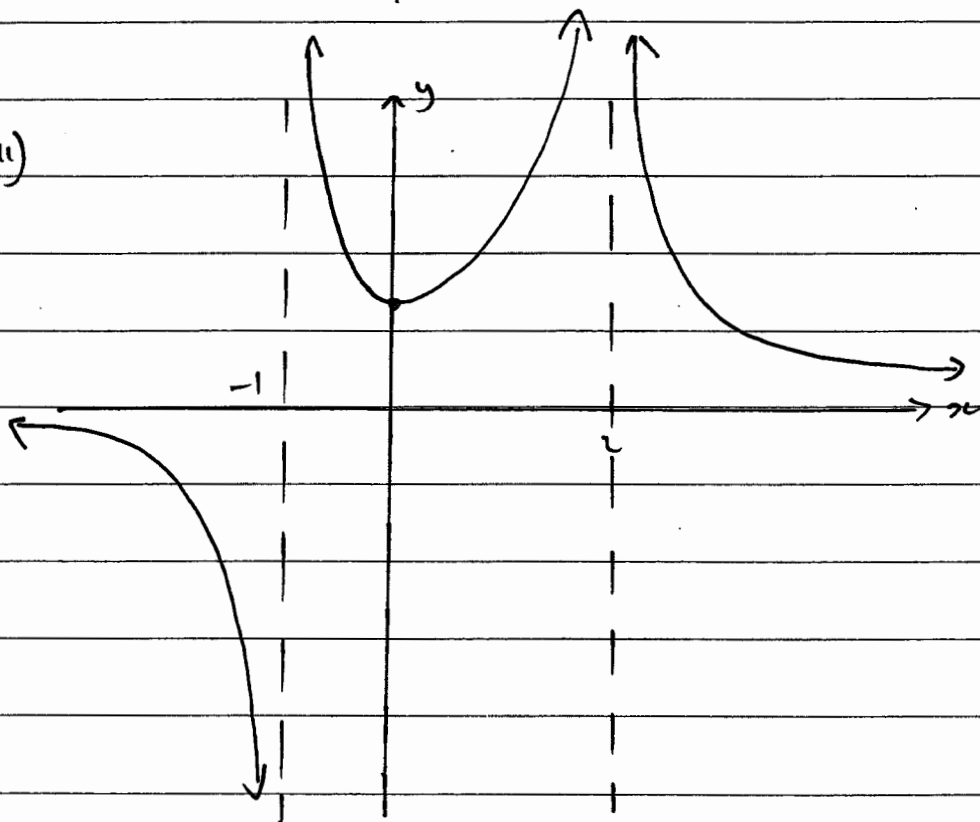
$$p = -2i$$

2. a.

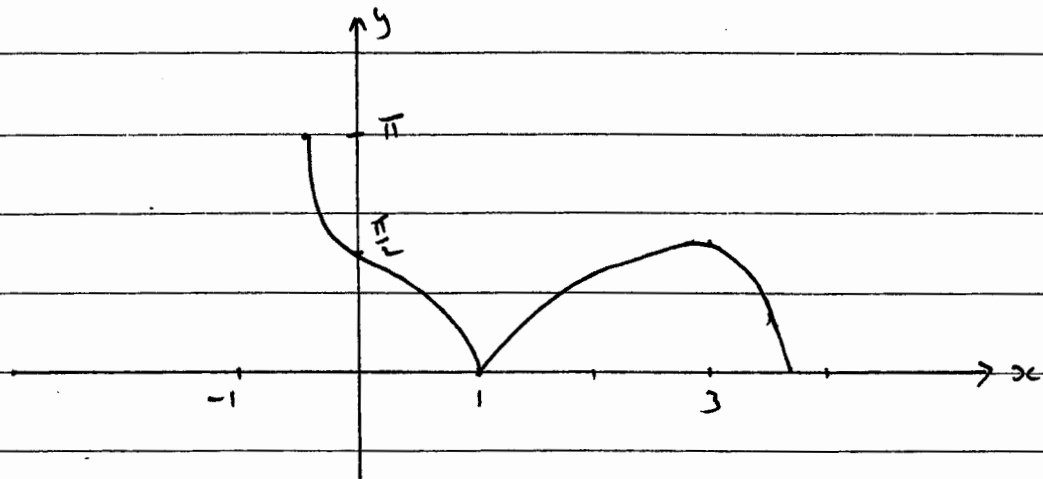
i)



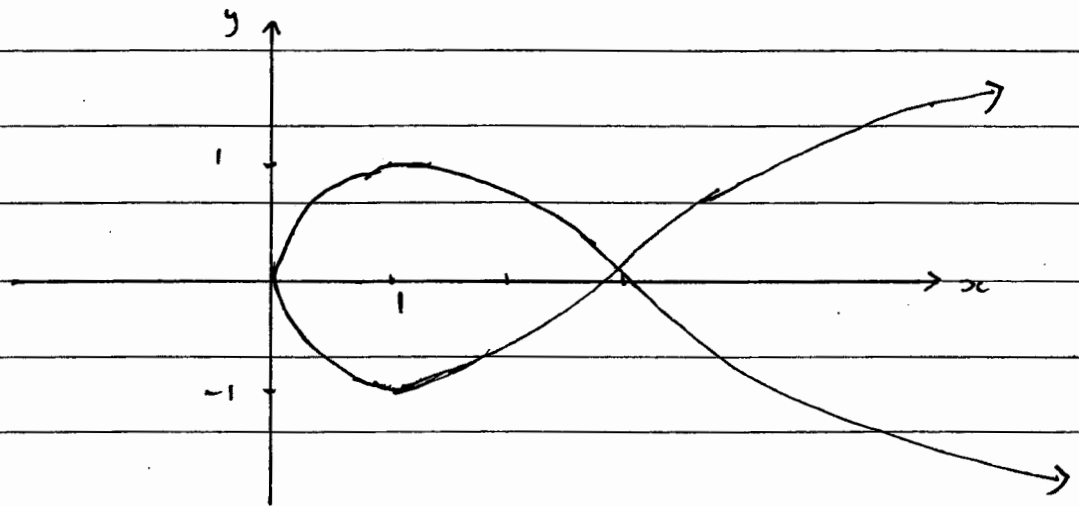
ii)



iii)



iv)



b) i) $e = \sqrt{2}$

$$\begin{aligned} \text{ii) } m &= \frac{\frac{2}{1} - \frac{2}{p}}{2q - 2p} \\ &= \frac{2p - 2q}{pq(q - p)} \\ &= \frac{-1}{pq} \end{aligned}$$

$$\therefore y - \frac{2}{p} = \frac{-1}{pq}(x - 2p)$$

$$pqy - 2q = -x + 2p$$

$$x + pqy = 2(p + q)$$

iii) sub $x=4, y=2$

$$4 + 2pq = 2(p+q)$$

$$2 + pq = p+q$$

$$pq = p+q-2$$

iv) midpoint $\left(p+q, \frac{p+q}{pq} \right)$

$$\therefore x = p+q, \quad y = \frac{p+q}{pq}$$

$$\therefore \text{ locus } y = \frac{p+q}{p+q-2}$$

$$y = \frac{x}{x-2}$$

c) $\int_0^{\frac{\pi}{6}} \sin 2x \sin 4x \, dx$

$$\sin 4x = 2 \sin 2x \cos 2x$$

$$= \int_0^{\frac{\pi}{6}} 2 \sin^2 2x \cos 2x \, dx$$

$$= \left[\frac{1}{3} \sin^3 2x \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{3} \left(\sin^3 \frac{\pi}{3} - \sin^3 0 \right)$$

$$= \frac{1}{3} \cdot \frac{3\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{8}$$

QUESTION 3

$$a) \int \frac{2x}{x^2+10x+26} dx$$

$$= \int \frac{2x+10}{x^2+10x+26} - \frac{10}{x^2+10x+26} dx$$

$$= \ln(x^2+10x+26) - \int \frac{10}{(x+5)^2+1} dx$$

$$= \ln(x^2+10x+26) - 10 \tan^{-1}(x+5) + C$$

$$b) \int \frac{1+\sin x}{\cos^2 x} dx$$

$$= \int \sec^2 x + \sec x \tan x dx$$

$$= \tan x + \sec x + C$$

$$c) P(x) = 2x^4 - 3x^2 - 2x + k$$

$$P'(x) = 8x^3 - 6x - 2$$

$$P''(x) = 24x^2 - 6$$

triple root is root of $P''(x) = 0$

$$24x^2 - 6 = 0$$

$$x^2 = \frac{1}{4}$$

$$x = \pm \frac{1}{2}$$

$$P'(\pm \frac{1}{2}) \neq 0 \therefore \text{triple root is } -\frac{1}{2}$$

$$\therefore 2(-\frac{1}{2})^4 - 3(-\frac{1}{2})^2 - 2(-\frac{1}{2}) + k = 0$$

$$\frac{1}{8} - \frac{3}{4} + 1 + k = 0$$

$$k = -\frac{3}{8}$$

d) i) required polynomial $\Rightarrow P\left(\frac{x}{z}\right) = 0$

$$\left(\frac{x}{z}\right)^3 - 6\left(\frac{x}{z}\right)^2 - 4\left(\frac{x}{z}\right) + 2 = 0$$

$$\frac{x^3}{8} - \frac{3x^2}{2} - 2x + 2 = 0$$

$$x^3 - 12x^2 - 16x + 16 = 0$$

ii) $y = \frac{1+x}{1-x}$

$$y - xy = 1 + x$$

$$y - 1 = x(y + 1)$$

$$x = \frac{y-1}{y+1}$$

$\therefore$ required polynomial $\Rightarrow P\left(\frac{y-1}{y+1}\right) = 0$

$$\left(\frac{y-1}{y+1}\right)^3 - 6\left(\frac{y-1}{y+1}\right)^2 - 4\left(\frac{y-1}{y+1}\right) + 2 = 0$$

$$(y-1)^3 - 6(y-1)^2(y+1) - 4(y-1)(y+1)^2 + 2(y+1)^3 = 0$$

coefficient of y^3 is $1 - 6 - 4 + 2 = -7$

coefficient of constant is $-1 - 6 + 4 + 2 = -1$

$\therefore$ constant term = $\frac{1}{7}$

Teacher's Name:

Student's Name/Nº:

$$e) \int_0^{\infty} 3ye^{-y}(1-e^{-y})^2 dy$$

$$\left[\begin{array}{ll} u = 3y & u' = 3 \\ v = \frac{1}{3}(1-e^{-y})^3 & v' = e^{-y}(1-e^{-y})^2 \end{array} \right]$$

$$= \left[y(1-e^{-y})^3 \right]_0^{\infty} - \int_0^{\infty} (1-e^{-y})^3 dy$$

$$= \left[y(1-e^{-y})^3 \right]_0^{\infty} - \int_0^{\infty} 1 - 3e^{-y} + 3e^{-2y} - e^{-3y} dy$$

$$= \left[y(1-e^{-y})^3 - y + 3e^{-y} + \frac{3}{2}e^{-2y} - \frac{1}{3}e^{-3y} \right]_0^{\infty}$$

$$= (0) - \left(-3 + \frac{3}{2} - \frac{1}{3} \right)$$

$$= \frac{11}{6}$$